2 Geometry and Measurement

The big idea. Geometry is the study of *shapes*; measurement attaches *numbers* to those shapes: lengths, angle measures, perimeters, areas and volumes. Almost every problem in this topic is solved with a handful of very simple ideas that are used over and over again:

- **Angles have a fixed "budget":** two supplementary angles always add up to 180°, the three angles of a triangle always add up to 180°, and a full turn is 360°. Knowing some of the angles, we can work out the rest.
- **Areas can be cut, rearranged and subtracted:** a complicated shape is split into familiar pieces (rectangles, triangles, trapezoids), or computed as "big shape minus the extra bits".
- **Ratios instead of numbers:** "what percentage is shaded" only requires comparing the shaded part with the whole shape – often without knowing any real sizes.
- **Shape does not change when turned or flipped:** rotating, reflecting or rolling a cube only changes *position*, never *size* or *shape*.
- **A box is built in layers:** volume = base area × height.

Whenever you meet a geometry problem, always **draw the figure and write the given numbers on it**, then ask yourself: "What do I know? What is missing? Which formula connects the two?".

① The coordinate plane

The coordinate system Oxy consists of two number lines that are perpendicular at the origin O:

- the horizontal axis Ox is called the **x-axis**; its positive direction points to the *right*;
- the vertical axis Oy is called the **y-axis**; its positive direction points *up*.

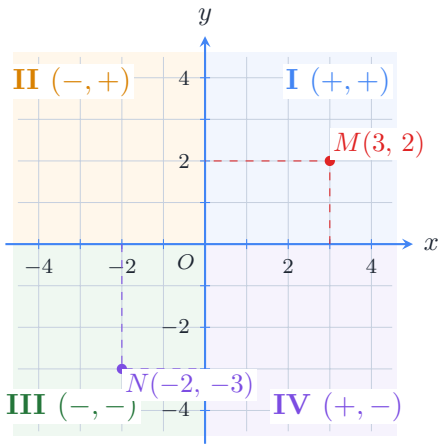
Each point M of the plane corresponds to exactly one pair of numbers (x_0, y_0), written $M(x_0, y_0)$:

- x_0 is the **x-coordinate**: drop a perpendicular from M to the x-axis; its foot is at the number x_0;
- y_0 is the **y-coordinate**: draw a perpendicular from M to the y-axis; its foot is at the number y_0.

The two axes divide the plane into four **quadrants**, numbered counterclockwise:

	Quadrant I	Quadrant II	Quadrant III	Quadrant IV
Signs of (x, y)	$(+, +)$	$(-, +)$	$(-, -)$	$(+, -)$

Distances along horizontal and vertical lines. Two points with the same y-coordinate (on the same horizontal line) are a distance apart equal to the difference of their x-coordinates (larger minus smaller). Similarly, two points with the same x-coordinate are a distance apart equal to the difference of their y-coordinates.



ThinkAlgo Tip. Read coordinates in the order "**across first, then up or down**": start at O, move right (positive) or left (negative) to get the x-coordinate; then move up (positive) or down (negative) to get the y-coordinate. For example, to reach $N(-2, -3)$: go 2 squares left, then 3 squares down.

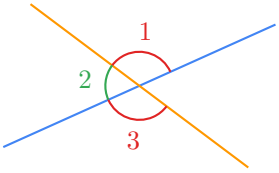
Common mistake. Swapping the two numbers is a very common mistake: $(3, -1)$ and $(-1, 3)$ are two *different* points (one in quadrant IV, the other in quadrant II). The x-coordinate is *always* written first.

Mini example. Let $A(1, 1)$, $B(5, 1)$, $C(5, 4)$. Since A and B have the same y-coordinate 1, $AB = 5 - 1 = 4$. Since B and C have the same x-coordinate 5, $BC = 4 - 1 = 3$. Angle B is a right angle (one side horizontal, one side vertical), so the area of triangle ABC is $\dfrac{4 \cdot 3}{2} = 6$ (square units).

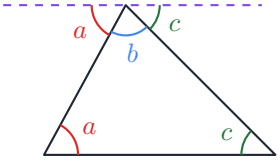
Note for parents. You can turn the coordinate plane into a "treasure hunt" on squared paper: you say "left 2, up 3", your child marks the point and writes its coordinates $(-2, 3)$. Then swap roles: your child gives directions and you draw. After a few rounds, the order "x first, y second" becomes automatic.

② Angles: supplementary, vertically opposite, angle sum of a triangle

- A **right angle** is 90°, a **straight angle** (half a turn) is 180°, a **full turn** is 360°.
- **Two adjacent supplementary angles** (sharing one arm, with the other two arms forming a straight line) add up to 180°.
- **Two vertically opposite angles** (each arm of one is the extension of an arm of the other) are **equal**.
- **The three angles of a triangle add up to** 180°**.** Hence: third angle = 180° − (sum of the two known angles).
- An **exterior angle of a triangle** (supplementary to an interior angle) equals **the sum of the two interior angles not adjacent to it**.
- In a right triangle, the two acute angles add up to 90°.



$\widehat{1}+\widehat{2}=180°$ (supplementary)
$\widehat{1}=\widehat{3}$ (vertically opposite)



$a+b+c=180°$

Why is it true? **Vertically opposite angles are equal.** In the left figure, $\widehat{1}+\widehat{2}=180°$ (supplementary) and $\widehat{3}+\widehat{2}=180°$ (also supplementary). Angles $\widehat{1}$ and $\widehat{3}$ are both supplementary to the same angle $\widehat{2}$, so $\widehat{1}=180°-\widehat{2}=\widehat{3}$.

The angles of a triangle add up to $180°$**.** Through the top vertex of the triangle, draw the line parallel to the base (dashed). For parallel lines, *alternate angles* are equal: the angle on the left at the top vertex equals a, and the angle on the right equals c. The three angles a, b, c at the top vertex together form a straight angle, so $a+b+c=180°$.

Exterior angle. The exterior angle at a vertex is supplementary to the interior angle there: exterior angle $=180°-$ interior angle. But $180°-$ interior angle is exactly the sum of the other two interior angles. So the exterior angle equals the sum of the two interior angles not adjacent to it.□

ThinkAlgo Tip. In "find the angle" problems, write the measure of every angle you find **on the figure** immediately. An angle you have just found is often the key to the next one (supplementary, vertically opposite, angle sum, ...).

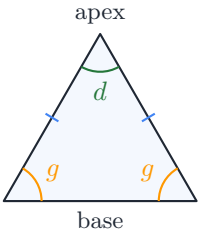
Common mistake. Do not confuse *vertically opposite* angles (equal) with *supplementary* angles (sum $180°$). Vertically opposite angles sit "across from each other" at the intersection point and share no arm; adjacent supplementary angles sit "next to each other" and share an arm.

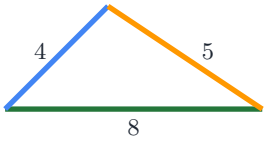
③ Isosceles triangles, equilateral triangles and the triangle inequality

- An **isosceles triangle** has two equal sides (the *legs*); the third side is the *base*, and the angle between the two legs is the *apex angle*.
- In an isosceles triangle, **the two base angles are equal**. So if the apex angle is d and each base angle is g, then

$$d+2g=180°, \qquad d=180°-2g, \qquad g=\frac{180°-d}{2}.$$

- A base angle is always **acute** (less than $90°$), because the two base angles are equal and their sum must be less than $180°$.
- An **equilateral triangle** has three equal sides and three equal angles, each $180°\div 3=60°$.
- **Triangle inequality:** in a triangle, each side is always *less* than the sum of the other two. To check whether three lengths form a triangle, it is enough to check: *longest side* $<$ *sum of the other two.*





$8 < 4 + 5$: a triangle exists

Why is it true? **Why are the base angles equal?** Let M be the midpoint of the base BC of the isosceles triangle ABC ($AB = AC$). Triangles ABM and ACM have $AB = AC$, $BM = CM$ and the common side AM, so they are congruent (side – side – side). Hence the corresponding angles $\widehat{B}$ and $\widehat{C}$ are equal.

Why does the triangle inequality hold? The shortest path between two points is the straight segment. Going from B to C "via" A (that is, travelling $BA + AC$) is always longer than going straight along BC. If the longest side were equal to or greater than the sum of the other two, the two short sides could not "reach each other": they would lie flat along the long side or fail to meet, and no triangle would be formed. □

Common mistake. When a problem only says "an isosceles triangle has *an angle* of 50°", we do **not yet know** whether it is the apex angle or a base angle. We must consider **both cases**:

- 50° is the apex angle: the base angles are $(180° - 50°) \div 2 = 65°$ and 65°;
- 50° is a base angle: the other base angle is also 50°, and the apex angle is $180° - 2 \cdot 50° = 80°$.

Only when the given angle is *greater than or equal to* 90° must it be the apex angle.

④ Rotations, reflections and rolling cubes

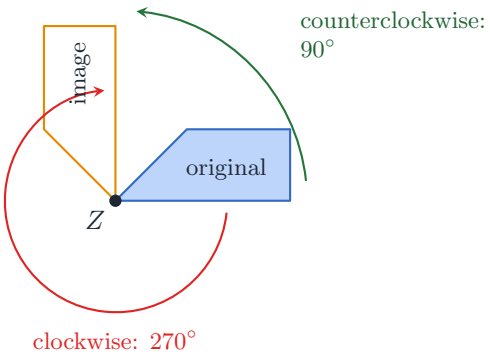
A **rotation** about a point Z (the *centre of rotation*) through an angle α in a given direction (clockwise or counterclockwise) sends each point M to a point M' such that $ZM' = ZM$ and $\widehat{MZM'} = \alpha$.

- Every point of the figure turns through **the same angle**, in the same direction; distances to the centre do not change.
- A rotation keeps the shape and size of the figure; the figure is only "turned", never "flipped over".
- Rotating α clockwise gives the same result as rotating $360° - \alpha$ counterclockwise. A 360° rotation brings the figure back to where it started.
- **How to measure the angle of rotation:** choose a segment starting at the centre Z on the original figure, find the matching segment on the new figure, and measure the angle that segment has swept through *in the given direction*.

A **reflection in a line** (flipping over the line, like a mirror) also keeps sizes, but **reverses the orientation** of the figure: the letter P seen in a mirror becomes a backwards P.

In the coordinate plane (centre of rotation at the origin O):

Rotation 90° counterclockwise	$(x, y) \to (-y, x)$
Rotation 180°	$(x, y) \to (-x, -y)$
Rotation 90° clockwise	$(x, y) \to (y, -x)$
Reflection in the x-axis / in the y-axis	$(x, y) \to (x, -y)$ / $(x, y) \to (-x, y)$



Why is it true? Why is a 270° clockwise rotation the same as a 90° counterclockwise one? A full turn is 360°. After going 270° one way, only $360° - 270° = 90°$ more would bring us back to the start; so the end position is exactly where we would arrive by going 90° the other way. In the figure: to turn the horizontal side (pointing right) into the vertical side (pointing up), we can rotate 90° counterclockwise, or rotate 270° clockwise (through pointing down, pointing left, and only then up). □

Rolling a cube. When a cube is rolled, each face is carried along unchanged. So a face with a painted pattern:

- may appear in another position (front, top, side, ...);
- may be *turned* by 90°, 180° or 270° within its own plane;
- **but never** becomes a mirror image, changes size, or "gains" extra paint.

ThinkAlgo Tip. For cube problems, "mark" one special feature of the pattern (for example: the right angle of the black triangle sits at *a corner* of the face) and check that feature in every option. Cutting out a paper square, drawing the pattern on it and turning it is also a very effective check.

⑤ Units of measurement and the number line

Units of length. Each unit is 10 times the next one:

$$1 \text{ km} = 10 \text{ hm}, \quad 1 \text{ hm} = 10 \text{ dam}, \quad 1 \text{ dam} = 10 \text{ m},$$

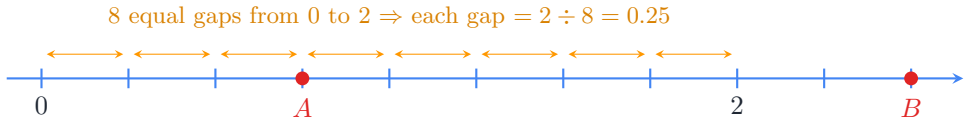
$$1 \text{ m} = 10 \text{ dm}, \quad 1 \text{ dm} = 10 \text{ cm}, \quad 1 \text{ cm} = 10 \text{ mm}.$$

The most used relations: $1 \text{ m} = 100 \text{ cm} = 1000 \text{ mm}$; $1 \text{ km} = 1000 \text{ m}$.

Units of area go up by a factor of 100: $1\ \text{m}^2 = 100\ \text{dm}^2 = 10\,000\ \text{cm}^2$.
Units of volume go up by a factor of 1000: $1\ \text{m}^3 = 1000\ \text{dm}^3$, $1\ \text{dm}^3 = 1000\ \text{cm}^3 = 1$ litre.
Conversion rule: from a *larger* unit to a *smaller* unit, **multiply**; from a *smaller* unit to a *larger* unit, **divide**.
Number line with equally spaced ticks. If we know the numbers at two ticks and count k *gaps* (not k ticks) between them, then

$$\text{value of one gap} = \frac{\text{difference of the two numbers}}{k}.$$

Why is it true? Why is $1\ \text{m}^2 = 10\,000\ \text{cm}^2$ and not $100\ \text{cm}^2$? One square metre is the area of a square with side 1 m, i.e. side 100 cm. That square can be divided into 100 rows of 100 squares of side 1 cm, so it contains $100 \times 100 = 10\,000$ squares of $1\ \text{cm}^2$. Likewise, a cube of side 1 dm = 10 cm consists of $10 \times 10 \times 10 = 1000$ small cubes of $1\ \text{cm}^3$. □



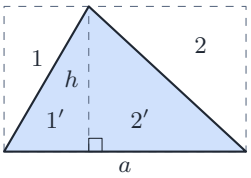
Mini example. On the number line above, A is three gaps from 0, so $A = 3 \cdot 0.25 = 0.75$; B is ten gaps from 0, so $B = 10 \cdot 0.25 = 2.5$.

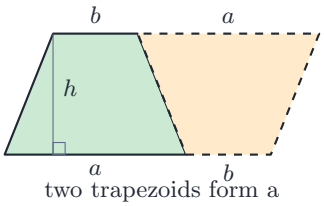
Common mistake. Counting *ticks* instead of *gaps*: from tick 0 to tick 2 in the figure above there are 9 ticks but only 8 gaps. Between two numbers there is always one gap fewer than ticks (like trees in a row: 9 trees have 8 spaces between them).

⑥ Perimeter and area of plane figures

Figure	Perimeter	Area
Rectangle (length a, width b)	$P = 2 \cdot (a + b)$	$S = a \cdot b$
Square (side a)	$P = 4 \cdot a$	$S = a \cdot a = a^2$
Triangle (base a, height h to that base)	sum of the three sides	$S = \frac{a \cdot h}{2}$
Parallelogram (base a, height h)	2·(sum of two adjacent sides)	$S = a \cdot h$
Trapezoid (parallel sides a, b; height h)	sum of the four sides	$S = \frac{(a + b) \cdot h}{2}$

The **height** of a triangle (parallelogram, trapezoid) is the segment drawn *perpendicular* from the vertex (or from the opposite side) to the line containing the base.
Working backwards: from $S = \frac{a \cdot h}{2}$ we get $h = \frac{2 \cdot S}{a}$ and $a = \frac{2 \cdot S}{h}$; from the perimeter of a square, side $= P \div 4$.





two trapezoids form a parallelogram with base $a + b$

Why is it true? **Area of a triangle.** Place the triangle inside the rectangle with the same base a and the same height h (left figure). The height splits the rectangle into two smaller rectangles; each smaller rectangle is cut by a diagonal into two equal triangles: the part inside the triangle ($1'$, $2'$) equals the part outside it (1, 2). So the triangle is exactly half of the rectangle: $S = \dfrac{a \cdot h}{2}$.
Area of a parallelogram. Cut off a right triangle at one end and slide it to the other end: we get a rectangle with the same base a and the same height h: $S = a \cdot h$.
Area of a trapezoid. Two identical trapezoids, one turned upside down, fit together into a parallelogram with base $a+b$ and height h (right figure). One trapezoid is half of it: $S = \dfrac{(a+b) \cdot h}{2}$.
□

ThinkAlgo Tip. Two triangles with **the same base** (or equal bases) and **the same height** have equal areas, even if they look very different. Consequence: the segment joining a vertex to the midpoint of the opposite side divides a triangle into two parts of equal area.

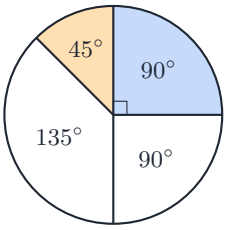
Common mistake.

- The height must be **perpendicular** to the base; a slanted side is *not* the height.
- Forgetting to divide by 2 in the area formulas of the triangle and the trapezoid.
- Mixing up units: perimeter is measured in cm, area in cm^2. A square with perimeter 36 cm has side 9 cm and area 81 cm^2 – not 36 cm^2.

⑦ Parts of an area: fractions, percentages, circles and rearranging

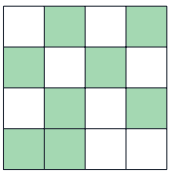
- If a figure is divided into n **equal** parts and m parts are shaded, the shaded part is $\dfrac{m}{n}$ of the figure, i.e. $\dfrac{m}{n} \cdot 100\%$.
- **Circles and sectors.** One turn around the centre is 360°. A sector with central angle $n°$ is $\dfrac{n}{360}$ of the area of the circle. For example: a semicircle corresponds to 180°, a quarter circle to 90° (a right angle at the centre).
- **Spinners:** the pointer stops at random, all directions being equally likely, so the chance of stopping in a region equals the fraction of the area (i.e. of the angle) of that region.
- **Cut and rearrange:** split the figure into familiar pieces and add their areas. **Complement:** required area = area of the big figure − area of the extra pieces.
- **Systematic counting of figures:** when asked how many figures (triangles, parallelograms, ...) appear in a diagram, sort them by *size* (made of 1 piece, 2 pieces, 3 pieces, ...) and count

each type so that nothing is missed or counted twice.



blue: $\frac{90}{360} = \frac{1}{4} = 25\%$

orange: $\frac{45}{360} = \frac{1}{8} = 12.5\%$



8 of 16 squares: $\frac{8}{16} = 50\%$

Why is it true? Why is a sector with angle n° equal to $\frac{n}{360}$ of the circle? Imagine cutting the circle into 360 tiny sectors, each with a central angle of 1°. These tiny sectors are identical (turning the circle by 1° about its centre makes one fit exactly onto the next), so they have equal areas, each $\frac{1}{360}$ of the circle. A sector with angle n° is made of n such tiny sectors. □

Common mistake. You may only count "m parts out of n" when the parts are **equal**. If the pieces have different sizes, reduce them to one kind of piece (for example by cutting them further into equal pieces) or compute the area of each piece.

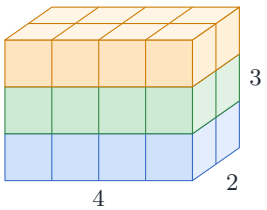
Note for parents. Cut a cake (or a paper circle) into 4, then into 8 equal parts so your child can see: 1 quarter = 90°, 1 eighth = 45°, and half of a quarter is an eighth. The idea "halving the angle halves the area" becomes very concrete.

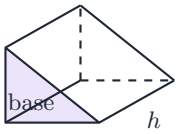
⑧ Solids: volume and surface area

- A **rectangular box** (rectangular prism) with dimensions a, b, c:

$$V = a \cdot b \cdot c, \qquad S_{\text{lateral}} = 2 \cdot (a + b) \cdot c, \qquad S_{\text{total}} = 2 \cdot (ab + bc + ca).$$

- A **cube** with side a: $V = a^3$, $S_{\text{total}} = 6 \cdot a^2$.
- A **right prism** (with a triangular, quadrilateral, ... base): $V = S_{\text{base}} \cdot h$, where h is the height of the prism (the length of a lateral edge).
- The number of 1 cm × 1 cm × 1 cm cubes needed to fill a box exactly equals the volume of the box in cm^3.





each layer $4 \cdot 2 = 8$ cubes, 3 layers: $8 \cdot 3 = 24$ cubes $V = S_{\text{base}} \cdot h$

Why is it true? **Why is** $V = a \cdot b \cdot c$**?** Stack 1 cm^3 cubes in layers. The bottom layer is a rectangle of a rows $\times$ b columns, containing $a \cdot b$ cubes (exactly the base area). Stack c such layers and we have $a \cdot b \cdot c$ cubes. The argument "each layer = base area, number of layers = height" works for every right prism, so $V = S_{\text{base}} \cdot h$. In particular, for a prism whose base is a right triangle with legs a, b: two such prisms fit together into an $a \times b \times h$ box, so each prism has volume $\dfrac{a \cdot b}{2} \cdot h$. $\square$

ThinkAlgo Tip. The surface area of a box is the total area of its 6 faces, which come in 3 pairs of equal opposite faces. When small blocks are joined into a big block, the faces that are "pressed together" are no longer on the surface, so the surface area of the big block is *less* than the total surface area of the small blocks.

Example 1

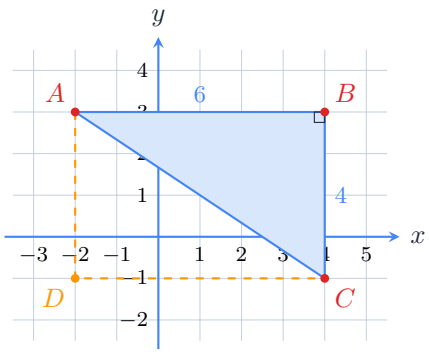
In the coordinate plane Oxy, consider the points $A(-2, 3)$, $B(4, 3)$ and $C(4, -1)$.

a) In which quadrant does each of the points A, B, C lie?

b) Find the lengths AB, BC and the area of triangle ABC.

c) Find the coordinates of the point D such that $ABCD$ is a rectangle.

Guided solution.



Step 1. Use the signs of the coordinates to find the quadrant

$A(-2, 3)$: negative x, positive y $\Rightarrow$ quadrant II. $B(4, 3)$: both positive $\Rightarrow$ quadrant I. $C(4, -1)$: positive x, negative y $\Rightarrow$ quadrant IV.

Step 2. Find the lengths of the horizontal and vertical sides

A and B have the same y-coordinate 3, so AB is horizontal and $AB = 4 - (-2) = 4 + 2 = 6$ (units). B and C have the same x-coordinate 4, so BC is vertical and $BC = 3 - (-1) = 3 + 1 = 4$.

Step 3. Find the area of the triangle

Side AB is horizontal and side BC is vertical, so angle B is a right angle. The area of a right triangle is half the product of the two legs:

$$S_{ABC} = \frac{AB \cdot BC}{2} = \frac{6 \cdot 4}{2} = 12 \text{ (square units)}.$$

Step 4. Find the point D

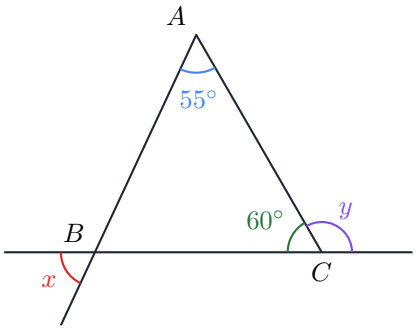
In rectangle $ABCD$, D is "opposite" B: D must be vertically in line with A (same x-coordinate -2) and horizontally in line with C (same y-coordinate -1). So $D(-2, -1)$, in quadrant III.

Check: Rectangle $ABCD$ has area $6 \cdot 4 = 24$, exactly twice the area of triangle ABC (the diagonal AC cuts the rectangle into two equal halves).

Answer: **a)** A: **II**, B: **I**, C: **IV**. **b)** $AB = 6$, $BC = 4$, $S = 12$. **c)** $D(-2, -1)$.

Example 2

In the figure below, triangle ABC has $\widehat{A} = 55°$, $\widehat{C} = 60°$. Sides AB and CB are extended beyond B; side BC is extended beyond C. Find the measures x and y.



Guided solution.

Analysis. Angle x lies *outside* the triangle, but it is *vertically opposite* to the angle $\widehat{ABC}$ of the triangle (the arms of x are the extensions of BC and of BA). So we find $\widehat{ABC}$ first.

Step 1. **Find angle B of the triangle**

Recall. The three angles of a triangle add up to 180°.

$\widehat{ABC} = 180° - (55° + 60°) = 180° - 115° = 65°$.

Step 2. **Find x**

Recall. Vertically opposite angles are equal.

Angle x and angle $\widehat{ABC}$ are vertically opposite, so $x = 65°$.

Step 3. **Find y**

Angle y is supplementary to angle $\widehat{ACB}$, so $y = 180° - 60° = 120°$.

Check: By the exterior angle property, y must equal the sum of the two non-adjacent interior angles: $55° + 65° = 120°$. It matches!

Answer: $x = 65°$, $y = 120°$.

Example 3

a) An isosceles triangle has an angle of 40°. Find the other two angles (consider all possibilities).

b) An isosceles triangle has an angle of 100°. Find the other two angles.

c) Which of the triples of lengths (3, 4, 8) and (3, 5, 7) forms a triangle? An isosceles triangle has two sides of lengths 4 cm and 9 cm. Find its perimeter.

Guided solution.

Step 1. Part a – two cases

Recall. An isosceles triangle has two equal base angles; the three angles add up to 180°.

Case 1: 40° *is the apex angle.* The base angles are equal and add up to $180° - 40° = 140°$, so each is $140° \div 2 = 70°$. The triangle has angles 40°, 70°, 70°.
Case 2: 40° *is a base angle.* The other base angle is also 40°; the apex angle is $180° - 40° - 40° = 100°$. The triangle has angles 40°, 40°, 100°.

Step 2. Part b – only one case

100° cannot be a base angle: otherwise the two base angles alone would add up to $200° > 180°$. So 100° is the apex angle, and each base angle is $(180° - 100°) \div 2 = 40°$.

Step 3. Part c – use the triangle inequality

Recall. Three lengths form a triangle exactly when the longest is less than the sum of the other two.

(3, 4, 8): $8 > 3 + 4 = 7$, *no* triangle. (3, 5, 7): $7 < 3 + 5 = 8$, a triangle exists.
An isosceles triangle with two sides 4 and 9 has its third side equal to 4 or to 9:

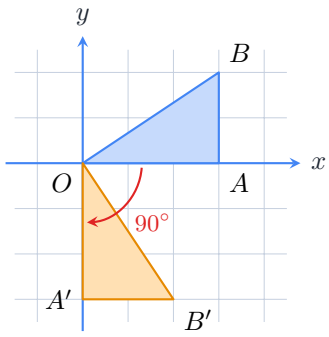
- (4, 4, 9): $9 > 4 + 4 = 8$, rejected;
- (4, 9, 9): $9 < 4 + 9 = 13$, fine. Perimeter $= 4 + 9 + 9 = 22$ cm.

Answer: a) 70° and 70°, or 40° and 100°. b) 40° and 40°. c) only (3, 5, 7); perimeter 22 cm.

Example 4

Triangle OAB with $O(0, 0)$, $A(3, 0)$, $B(3, 2)$ is rotated about the origin O *clockwise* and becomes triangle $OA'B'$ with $A'(0, -3)$, $B'(2, -3)$. What is the angle of rotation? If we rotate counterclockwise instead, through what angle must we turn to get the same result?

Guided solution.



Step 1. Choose a segment starting at the centre of rotation

Segment OA lies along the ray Ox, pointing *right*. Its image is OA' with $A'(0, -3)$, lying on the y-axis and pointing *down*.

Step 2. Measure the angle swept clockwise

A clock hand going from the "3 o'clock" direction (right) to the "6 o'clock" direction (down) sweeps

a right angle. So the clockwise angle of rotation is $90°$.

Step 3. Convert to the other direction

Recall. Rotating α clockwise gives the same result as rotating $360° - \alpha$ counterclockwise.

Counterclockwise: $360° - 90° = 270°$.

Check: By the table in section 4, a $90°$ clockwise rotation sends (x, y) to $(y, -x)$: $A(3, 0) \to (0, -3) = A'$ and $B(3, 2) \to (2, -3) = B'$. Correct.

Answer: $90°$ clockwise, equivalent to $270°$ counterclockwise.

Example 5

On a number line, the ticks are equally spaced. The first tick is at 0, the seventh tick is at 3 (that is, six gaps away from 0). Point M is at the tick three gaps from 0, point N at the tick nine gaps from 0.

a) Find the numbers at M and N, and compute $M \div N$.

b) If the unit on the line is the metre, convert M and N to centimetres.

Guided solution.



Step 1. Find the value of one gap

From 0 to 3 there are 6 equal gaps, so each gap is worth $3 \div 6 = 0.5$.

Step 2. Find M and N

M is three gaps from 0: $M = 3 \cdot 0.5 = 1.5$. N is nine gaps from 0: $N = 9 \cdot 0.5 = 4.5$.

Step 3. Compute the quotient

$M \div N = 1.5 \div 4.5 = \dfrac{15}{45} = \dfrac{1}{3} \approx 0.33$. (Quick view: N is 3 times as far from 0 as M, so $M \div N = \frac{1}{3}$.)

Step 4. Convert units

Recall. 1 m = 100 cm; from a larger unit to a smaller unit, multiply.

$1.5 \text{ m} = 1.5 \cdot 100 \text{ cm} = 150 \text{ cm}$; $4.5 \text{ m} = 450 \text{ cm}$.

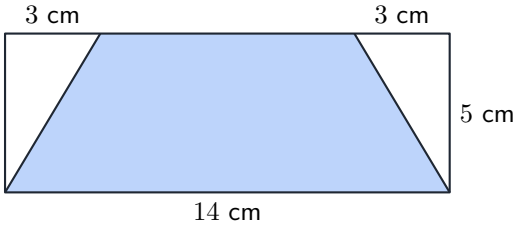
Answer: a) $M = 1.5$; $N = 4.5$; $M \div N = \frac{1}{3}$. b) 150 cm and 450 cm.

Example 6

A rectangle is 14 cm long and 5 cm wide. On its top side, two points are marked, each 3 cm from a top corner; they are joined to the two bottom corners to form the shaded trapezoid (see figure).

a) Find the area of the trapezoid in two ways.

b) A triangle has area 24 cm^2 and a base of length 8 cm. Find the height to that base.



Guided solution.

Step 1. **Method 1 – the trapezoid formula**

Recall. $S_{\text{trapezoid}} = \dfrac{(a+b)\cdot h}{2}$, where a, b are the parallel sides and h is the height.

Longer parallel side $a = 14$ cm. Shorter parallel side $b = 14 - 3 - 3 = 8$ cm. The height equals the width of the rectangle: $h = 5$ cm.

$$S = \frac{(14+8)\cdot 5}{2} = \frac{22\cdot 5}{2} = \frac{110}{2} = 55 \text{ (cm}^2\text{)}.$$

Step 2. **Method 2 – big shape minus the extra pieces**

Area of the rectangle: $14\cdot 5 = 70$ (cm^2). The two unshaded parts are right triangles, each with legs 3 cm and 5 cm and area $\dfrac{3\cdot 5}{2} = 7.5$ (cm^2). Area of the trapezoid: $70 - 2\cdot 7.5 = 70 - 15 = 55$ (cm^2).

Step 3. **Part b – find the height by working backwards**

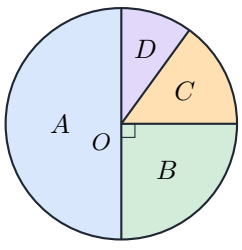
From $S = \dfrac{a\cdot h}{2}$ we get $h = \dfrac{2\cdot S}{a} = \dfrac{2\cdot 24}{8} = \dfrac{48}{8} = 6$ (cm).

Check: $\dfrac{8\cdot 6}{2} = 24$. Correct.

Answer: a) 55 cm^2. b) 6 cm.

Example 7

A circular spinner with centre O is divided into four regions A, B, C, D: region A is a semicircle, region B is a quarter circle, and the rest is split into regions C and D so that the central angle of C is one and a half times the central angle of D. What percentage of the spinner's area does each region take up? About what percentage of spins will stop in region C?



Guided solution.

Step 1. **Central angles of regions A and B**

The whole circle corresponds to 360°. Region A (a semicircle) corresponds to 180°, region B (a

quarter) to 90°.

Step 2. Central angles of regions C and D

The remainder: $360° - 180° - 90° = 90°$. We must split 90° into two parts with C one and a half times D; that is, if D is 2 equal parts then C is 3 such parts. Together 5 parts correspond to 90°, so one part is $90° \div 5 = 18°$. Hence the angle of D is $2 \cdot 18° = 36°$ and the angle of C is $3 \cdot 18° = 54°$.

Step 3. Convert to percentages

> **Recall.** A sector with central angle $n°$ is $\dfrac{n}{360}$ of the circle.

A: $\dfrac{180}{360} = 50\%$; B: $\dfrac{90}{360} = 25\%$; C: $\dfrac{54}{360} = 0.15 = 15\%$; D: $\dfrac{36}{360} = 0.1 = 10\%$.

Check: $50\% + 25\% + 15\% + 10\% = 100\%$, and $54° = 1.5 \cdot 36°$. Correct.

Answer: A: 50%, B: 25%, C: 15%, D: 10%; about 15% of spins stop in region C.

Example 8

a) How many cubes with edge 1 cm are needed to fill a 5 cm × 3 cm × 2 cm box? Find the total surface area of the box.

b) A right prism has a base that is a right triangle with legs 3 cm and 4 cm, and the prism is 10 cm high. Find its volume. If a prism with the same base has volume 90 cm^3, what is its height?

Guided solution.

Step 1. Part a – count layer by layer

> **Recall.** $V_{\text{box}} = a \cdot b \cdot c$; $S_{\text{total}} = 2 \cdot (ab + bc + ca)$.

The bottom layer has $5 \cdot 3 = 15$ cubes; there are 2 layers, so we need $15 \cdot 2 = 30$ cubes (volume 30 cm^3).

The three pairs of opposite faces have areas $5 \cdot 3 = 15$, $3 \cdot 2 = 6$, $5 \cdot 2 = 10$ (cm^2), so $S_{\text{total}} = 2 \cdot (15 + 6 + 10) = 2 \cdot 31 = 62$ (cm^2).

Step 2. Part b – base area times height

> **Recall.** $V_{\text{prism}} = S_{\text{base}} \cdot h$.

The base is a right triangle: $S_{\text{base}} = \dfrac{3 \cdot 4}{2} = 6$ (cm^2). Volume: $V = 6 \cdot 10 = 60$ (cm^3).

With volume 90 cm^3: $h = V \div S_{\text{base}} = 90 \div 6 = 15$ (cm).

Check: $6 \cdot 15 = 90$. Correct.

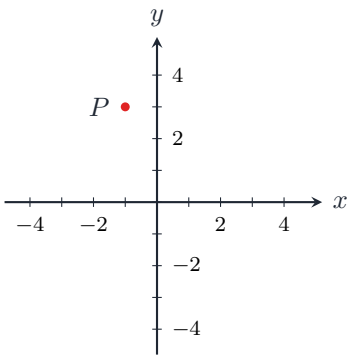
Answer: a) 30 cubes; 62 cm^2. b) 60 cm^3; 15 cm.

Note for parents. After each example, ask your child to *say in words* the formula just used and *why* it is true (for example: "a triangle is half of the rectangle with the same base and height"). Understanding the "why" means less rote memorisation and fewer mistakes on unfamiliar figures.

Core practice

Problem 1

In the coordinate plane below, each tick on the two axes represents 1 unit. Which pair of numbers could be the coordinates of point P?



(A) (1, 3) (B) (1, −3) (C) (−3, 1) (D) (3, −1) (E) (−1, 3)

Your work:

Problem 2

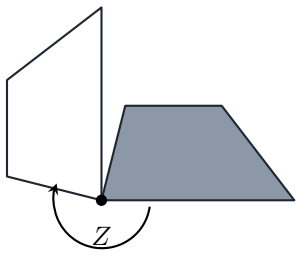
Two angles of a triangle measure 25° and 70°. The third angle of the triangle measures

(A) 85° (B) 105° (C) 65° (D) 95° (E) 75°

Your work:

Problem 3

Rotating the shaded quadrilateral about point Z *clockwise* (in the direction of the arrow in the figure) produces the unshaded quadrilateral. The angle of rotation is approximately

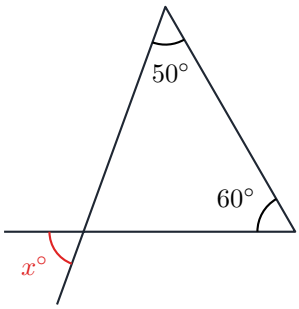


(A) 180° (B) 270° (C) 360° (D) 45° (E) 135°

Your work:

Problem 4

In the figure, two sides of the triangle are extended as shown. The value of x is

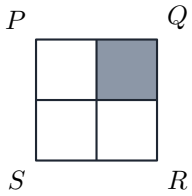


(A) 110 (B) 50 (C) 10 (D) 60 (E) 70

Your work:

Problem 5

Square $PQRS$ is divided into four equal smaller squares as shown. What percentage of the area of square $PQRS$ is shaded?

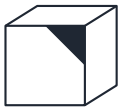


(A) 33% (B) 40% (C) 25% (D) 50% (E) 15%

Your work:

Problem 6

A cube has part of *exactly one face* painted black as shown (the paint is a right triangle at a corner of the face, whose two legs are half the edge of the cube). The other five faces are not painted. After the cube is rolled, which of the following could be the same cube?



(A) 

(B) 

(C) 

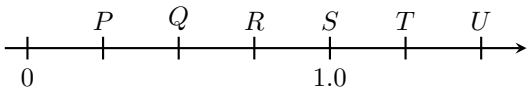
(D) 

(E) 

Your work:

Problem 7

On a number line the ticks are equally spaced. The numbers P, Q, R, S, T, U are marked as shown. Which of the following is closest to the value of $R \div U$?



(A) 0.25 (B) 0.50 (C) 0.75 (D) 1.25 (E) 1.50

Your work:

Problem 8

One angle of an isosceles triangle measures 50°. The other two angles of this triangle could be

(A) 50° and 90° (B) 40° and 50° (C) 50° and 80°

(D) 30° and 100° (E) 60° and 70°

Your work:

Problem 9

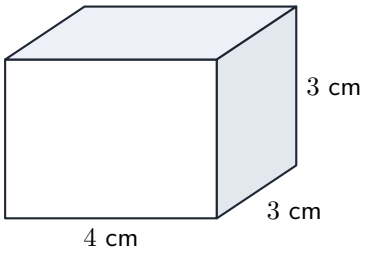
Minh measures the length of his desk as 3.5 m. How many centimetres is that?

(A) 350 (B) 30.5 (C) 3.05 (D) 3.5 (E) 305

Your work:

Problem 10

An wants to build the solid rectangular box shown from small cubes measuring 1 cm × 1 cm × 1 cm. How many small cubes does An need?



(A) 10 (B) 12 (C) 33 (D) 66 (E) 36

Your work:

Problem 11

A square garden has a perimeter of 36 m. The area of the garden, in m^2, is

(A) 24 (B) 81 (C) 36 (D) 1296 (E) 324

Your work:

Problem 12

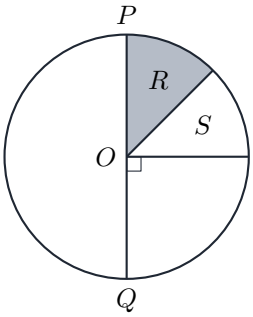
Two isosceles triangles each have at least one angle of 70°. In the first triangle, each of the two remaining angles measures an even number of degrees. In the second triangle, each of the two remaining angles measures an odd number of degrees. Let S be the sum of the two equal angles of the first triangle and T the sum of the two equal angles of the second triangle. The value of $S + T$ is

(A) 280° (B) 250° (C) 220° (D) 200° (E) 300°

Your work:

Problem 13

On the circular spinner with centre O shown, PQ is a diameter, the horizontal radius is perpendicular to PQ at O, and the sectors R and S have equal areas. About what percentage of spins will stop in the shaded region?

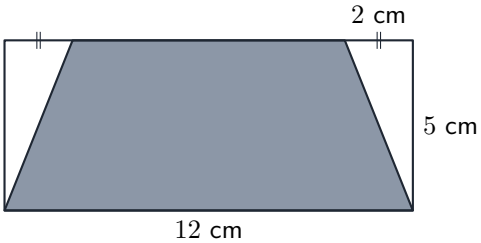


(A) 50% (B) 22.5% (C) 25% (D) 45% (E) 12.5%

Your work:

Problem 14

In the rectangle below (12 cm long, 5 cm wide), the two segments marked alike on the top side are each 2 cm long. The area of the shaded region is

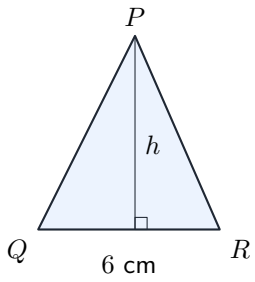


(A) 60 cm^2 (B) 20 cm^2 (C) 30 cm^2 (D) 40 cm^2 (E) 50 cm^2

Your work:

Problem 15

Triangle PQR has area 27 cm^2 and base $QR = 6$ cm. The height h of triangle PQR (to the base QR) is



(A) 9 cm (B) 18 cm (C) 4.5 cm (D) 2.25 cm (E) 7 cm

Your work:

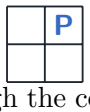
Problem 16

A rectangular garden has a length that is 6 m more than *twice* its width. The perimeter of the garden is 120 m. How many metres wide is the garden?

(A) 8 m (B) 18 m (C) 27 m (D) 38 m (E) 22 m

Your work:

Problem 17

The letter P is written in a 2×2 grid of squares as shown: 

. A sequence of rotations about the centre of the grid and reflections in the two lines through the centre of the grid (the horizontal line and the vertical line) is carried out, and the result is: 
.

If *exactly the same sequence of transformations* is applied to the grid 

, which picture do we get?

(A) 

(B) 

(C) 

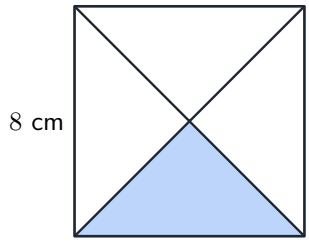
(D) 

(E) 

Your work:

Problem 18

In a square with side 8 cm, the two diagonals are drawn. What is the area of the shaded region?

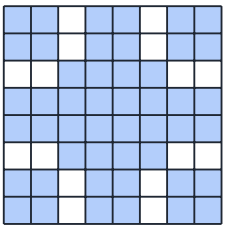


(A) 4 cm^2 (B) 8 cm^2 (C) 16 cm^2 (D) 56 cm^2 (E) 64 cm^2

Your work:

Problem 19

The grid below is made of identical small squares. What percentage of the area of the whole grid is shaded?

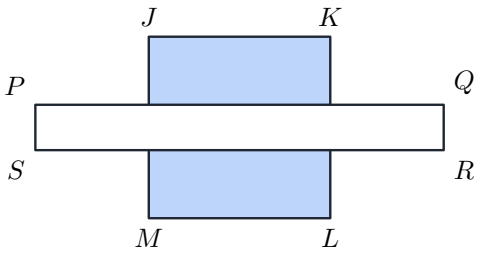


(A) 84% (B) 80% (C) 90% (D) 75% (E) 66%

Your work:

Problem 20

$JKLM$ is a square and $PQRS$ is a rectangle; they overlap as shown in the diagram. Given that JK is parallel to PQ, $JK = 8$ and $PS = 2$, find the total area of the shaded regions.



(A) 32 (B) 16 (C) 56 (D) 48 (E) 62

Your work:

Problem 21

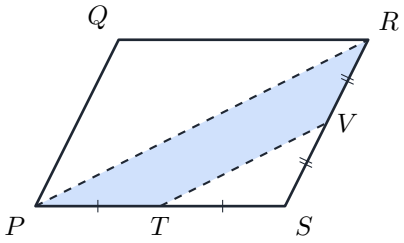
A triangle can be built with side lengths 4, 5, 8, but *no* triangle can be built with side lengths 4, 5, 10. Using only the lengths 2, 3, 5, 7, 11 (each length may be used more than once), how many different triangles with *exactly two equal sides* can be built?

(A) 8 (B) 5 (C) 20 (D) 10 (E) 14

Your work:

Problem 22

Parallelogram $PQRS$ has area 40. Points T and V are the midpoints of sides PS and RS respectively. Find the area of quadrilateral $PRVT$.

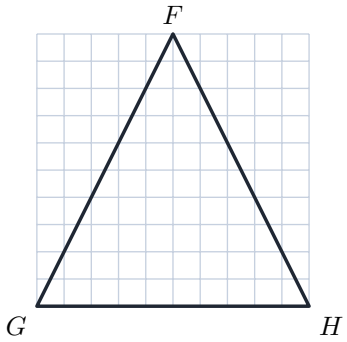


(A) 10 (B) 12 (C) 15 (D) 16 (E) 18

Your work:

Problem 23

On a 10×10 grid of squares (each square has side 1), triangle FGH is drawn as shown. Point P can be any of the 41 points where gridlines cross *inside* (not on the sides of) triangle FGH. For each position of P we get exactly three triangles: FPG, GPH, HPF. Of all these 123 triangles, how many have an area *exactly half* the area of triangle FGH?



(A) 5 (B) 7 (C) 3 (D) 11 (E) 9

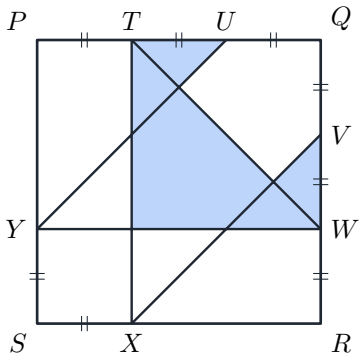
Your work:

Problem 24

Points T, U, V, W, X, Y lie on the sides of square $PQRS$ as shown, with

$$PT = TU = UQ = QV = VW = WR = XS = SY.$$

What fraction of the area of square $PQRS$ is shaded?

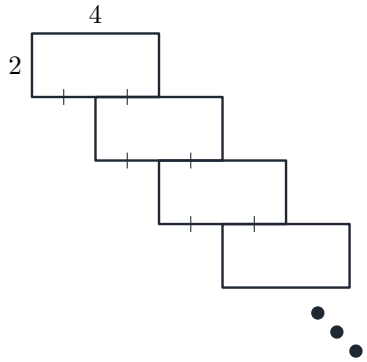


(A) $\frac{5}{18}$ (B) $\frac{1}{3}$ (C) $\frac{2}{9}$ (D) $\frac{1}{4}$ (E) $\frac{1}{6}$

Your work:

Problem 25

Rectangles measuring 4×2 are arranged by the following rule: from the second rectangle on, the top-left vertex of each rectangle is placed exactly at the midpoint of the bottom side of the rectangle just above it (as shown). When 10 rectangles in total are arranged in this way, what is the perimeter of the whole figure?



(A) 48 (B) 64 (C) 90 (D) 84 (E) 100

Your work:

Problem 26

A quadrilateral has vertices $A(7, 6)$, $B(-5, 1)$, $C(-2, -3)$ and $D(10, 2)$ in the coordinate plane. The area of the quadrilateral (in square units) is

(A) 60 (B) 63 (C) 67 (D) 70 (E) 72

Your work:

Problem 27

The figure below is made of 8 identical small parallelograms joined together. Including these 8 small parallelograms, how many parallelograms are there in the figure altogether?

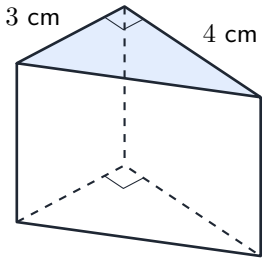


(A) 29 (B) 30 (C) 26 (D) 27 (E) 28

Your work:

Problem 28

A right triangular prism has volume 120 cm^3. Its base is a right triangle whose two legs are 3 cm and 4 cm (as shown). What is the height of the prism, in centimetres?



(A) 12 (B) 20 (C) 10 (D) 16 (E) 8

Your work:

Problem 29

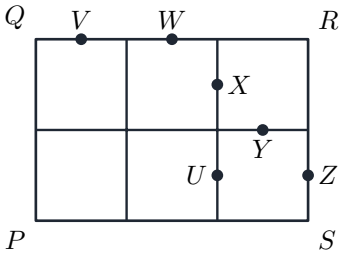
Small cubes with edge 1 are used to build a rectangular box of size $8 \times 8 \times n$ (n a positive integer). Let A be the surface area of the box and let B be the total surface area of all the small cubes (computed for each cube separately and then added up). Find the sum of all values of n for which $\frac{B}{A}$ is an integer.

(A) 86 (B) 90 (C) 70 (D) 78 (E) 96

Your work:

Problem 30

Rectangle $PQRS$ is made up of six identical squares. Points U, V, W, X, Y, Z are midpoints of sides of the squares, as shown. Which of the following triangles has the greatest area?



(A) PVU (B) PXZ (C) PVX (D) PYS (E) PQW

Your work:

Extension and challenge

Problem 31

In the coordinate plane Oxy, consider the point $A(-3, 2)$.

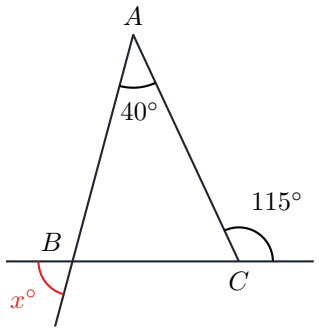
a) Let B be the reflection of A in the x-axis, C the reflection of A in the y-axis, and D the reflection of A in the origin O. Find the coordinates of B, C, D and state the quadrant of each point.

b) The four points A, C, D, B are the vertices of a rectangle. Find the perimeter and the area of that rectangle.

Your work:

Problem 32

In the figure, triangle ABC has $\widehat{BAC} = 40°$; side BC is extended in both directions and side AB is extended beyond B. The exterior angle at vertex C is 115°. The value of x is



(A) 65 (B) 75 (C) 105 (D) 115 (E) 140

Your work:

Problem 33

a) In an isosceles triangle, the apex angle is 4 times each base angle. Find the angles of the triangle.

b) An isosceles triangle has an angle of 36°. Find all possibilities for the other two angles. For each possibility, find the difference between the largest and the smallest angle of the triangle.

c) Is there an isosceles triangle with a base angle of 92°? Why?

Your work:

Problem 34 *

In the coordinate plane, consider the point $A(4, 1)$.

a) Rotating A about the origin O through 90° counterclockwise gives the point B; rotating B a further 90° counterclockwise gives C; rotating C a further 90° counterclockwise gives E. Find the coordinates of B, C, E.

b) Rotating A about O through 270° *clockwise* gives which of these three points?

c) The quadrilateral $ABCE$ is a square. Find its area.

Your work:

Problem 35

Fill in the blanks:

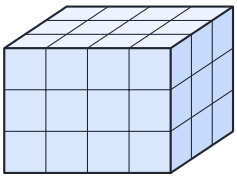
a) 2.4 km = ... m; b) 75 mm = ... cm; c) 3 m^2 = ... cm^2; d) 2500 cm^3 = ... litres.

e) A rectangular board is 1.2 m long and 50 cm wide. Find the area of the board in cm^2 and in m^2.

Your work:

Problem 36 *

An built a 4 cm × 3 cm × 3 cm box from 36 small cubes of edge 1 cm (as in Problem 10). An paints all the outside faces of the box and then takes the small cubes apart. How many small cubes have exactly 3 painted faces? Exactly 2? Exactly 1? No painted face?



Your work:

Problem 37

A rectangle has the same perimeter as a square of area 81 cm^2. The length of the rectangle is twice its width. The area of the rectangle is

(A) 81 cm^2 (B) 72 cm^2 (C) 54 cm^2 (D) 36 cm^2 (E) 64 cm^2

Your work:

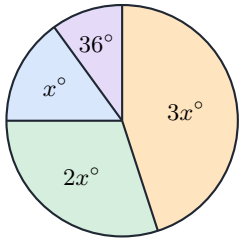
Problem 38

A circular spinner is divided into four regions with central angles 36°, $x°$, $2x°$ and $3x°$ (see figure).

a) Find x.

b) What percentage of the spinner's area does each region take up?

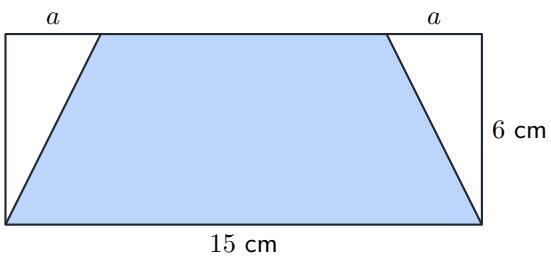
c) About what percentage of spins will *not* stop in the largest region?



Your work:

Problem 39

A rectangle is 15 cm long and 6 cm wide. On its top side, two segments of the same length a cm are taken at the two ends, and the two right triangles at the top corners are cut off, leaving the shaded trapezoid shown. The area of the trapezoid is 72 cm^2. Find a and the length of the shorter parallel side of the trapezoid.



Your work:

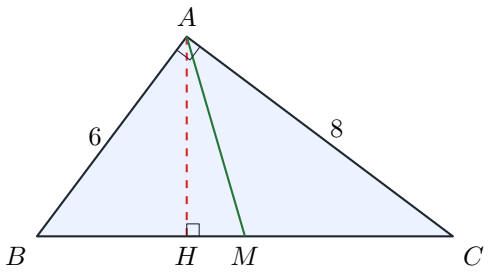
Problem 40

Triangle ABC has a right angle at A, with $AB = 6$ cm, $AC = 8$ cm and $BC = 10$ cm.

a) Find the area of triangle ABC.

b) Find the length of the height AH drawn from A to side BC.

c) Let M be the midpoint of BC. Find the area of triangle ABM.



Your work:

Problem 41 ⋆

(Extension of Problem 16.) We only consider rectangles whose two side lengths are *positive integers* (in cm); a square also counts as a rectangle.

a) Find all rectangles whose area (in cm^2) is numerically *equal* to their perimeter (in cm).

b) Find all rectangles whose area is numerically *twice* their perimeter.

Your work:

Problem 42 ⋆

(Extension of Problem 17.) The letter P is written in the top-right cell of a 2×2 grid: 

. We may use three transformations, as many times as we like and in any order:

- Q: rotate the grid 90° clockwise about its centre;
- D: reflect the grid in the *vertical* line through the centre;
- N: reflect the grid in the *horizontal* line through the centre.

a) Draw the result of doing D and then N. Can that result be obtained using only Q?

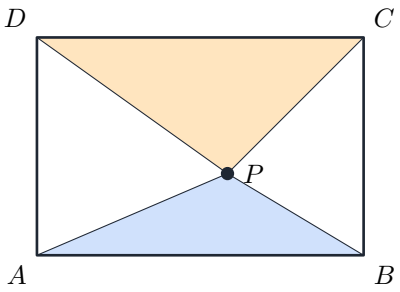
b) Show that doing D and then Q has the same effect as a reflection in one of the *diagonals* of the grid. Which diagonal?

c) How many different pictures can be obtained altogether (including the starting picture)? Explain why there cannot be more.

Your work:

Problem 43 *

(Extension of Problem 18.) Rectangle $ABCD$ has $AB = 12$ cm and $BC = 8$ cm. Point P lies inside the rectangle. (We write S_{XYZ} for the area of triangle XYZ.)



a) Prove that wherever P is, we always have $S_{PAB} + S_{PCD} = 48$ cm^2.

b) Given $S_{PAB} = 18$ cm^2 and $S_{PBC} = 20$ cm^2, find S_{PCD}, S_{PDA} and the distances from P to the four sides of the rectangle.

c) Is there a point P inside the rectangle with $S_{PAB} = 20$, $S_{PBC} = 30$ and $S_{PCD} = 30$ (cm^2)?

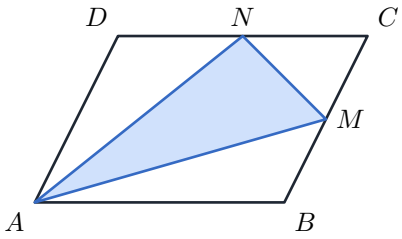
Your work:

Problem 44 *

(Extension of Problem 22.) Parallelogram $ABCD$ has area 48 cm^2.

a) Let M and N be the midpoints of BC and CD respectively. Find the area of triangle AMN.

b) Keep N as the midpoint of CD, but take M on side BC with $BM = 3\,MC$. Find the area of triangle AMN.



Your work:

Problem 45 *

(Extension of Problem 21.) Consider triangles whose three side lengths are positive integers and whose perimeter is 15 (two triangles with the same three side lengths count as one).

a) Let c be the longest side. Prove that $5 \le c \le 7$.

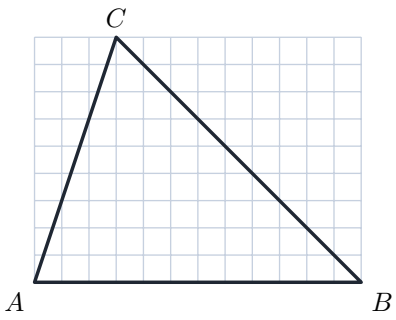
b) How many such triangles are there? List them.

c) How many of them are isosceles? How many have *exactly* two equal sides?

Your work:

Problem 46 **

(Extension of Problem 23.) On a square grid (each square has side 1), triangle ABC has vertices $A(0,0)$, $B(12,0)$, $C(3,9)$. Exactly 43 points where gridlines cross lie *inside* the triangle (not on its sides). For each such point P we get three triangles PAB, PBC, PCA.



a) Find the area of triangle ABC.

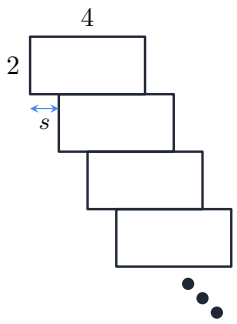
b) Find the point P for which the three triangles PAB, PBC, PCA have equal areas.

c) Of all the $3 \cdot 43 = 129$ triangles, how many have an area of exactly $\frac{1}{3}$ of the area of triangle ABC? For how many positions of P is at least one of the three triangles like that?

Your work:

Problem 47 *

(Extension of Problem 25.) Again we arrange 4×2 rectangles in a "staircase" as in Problem 25, but this time the top-left vertex of each rectangle (from the second one on) is placed on the bottom side of the rectangle just above it, at distance s from that rectangle's bottom-left vertex, where s is 1, 2 or 3. (Problem 25 is the case $s = 2$.)



a) For $s = 1$ and 10 rectangles, find the perimeter of the whole figure.

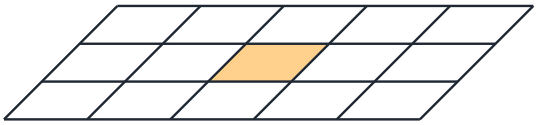
b) Find a formula for the perimeter in terms of the number of rectangles n and the distance s.

c) Can we choose $s \in \{1, 2, 3\}$ and a number of rectangles n so that the perimeter is exactly 100? Find all the ways.

Your work:

Problem 48 *

(Extension of Problem 27.) The figure below consists of 15 identical small parallelograms arranged in 3 rows of 5. The small parallelogram in the very middle is shaded.



a) How many parallelograms are there in the figure altogether?

b) How many parallelograms contain the shaded small parallelogram?

c) How many parallelograms are made of *exactly* 4 small parallelograms?

Your work:

Problem 49 ★★

(Extension of Problems 28 and 29.) A $4 \times 5 \times 6$ rectangular box is built from 120 small cubes with edge 1. All the outside faces of the box are painted, and then the small cubes are separated.

a) How many small cubes have exactly 3 painted faces? exactly 2? exactly 1? no painted face at all?

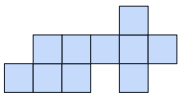
b) What is the total number of painted small faces (1×1 squares)? Explain why the result equals the surface area of the box.

c) An $n \times n \times n$ cube ($n \geq 3$) is painted and separated in the same way. Find n such that the number of small cubes with exactly 1 painted face equals the number of small cubes with no painted face.

Your work:

Problem 50 ★★

(Extension of Problems 19 and 25.) Ten unit squares (side 1) are joined to form a flat shape such that: adjacent squares share a whole side, and the shape is "in one piece" (from any square one can reach any other square through shared sides). The perimeter of the shape is the total length of the square sides that are not covered by another square.



an example

a) Show that the perimeter is always an even number.

b) What is the greatest possible perimeter?

c) What is the smallest possible perimeter?

d) Which numbers can be the perimeter of such a shape?

Your work: